

Vulnerability Assessment Report

09th February 2026

Scenario

You are a newly hired cybersecurity analyst for an e-commerce company. The company stores information on a remote database server, since many of the employees work remotely from locations all around the world. Employees of the company regularly query, or request, data from the server to find potential customers. The database has been open to the public since the company's launch three years ago. As a cybersecurity professional, you recognize that keeping the database server open to the public is a serious vulnerability.

You are tasked with completing a vulnerability assessment of the situation to communicate the potential risks to decision makers at the company. You must create a written report that explains how the vulnerable server is a risk to business operations and how it can be secured.

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

Assessment covers from March 2026 to May 2026. Guide used is [NIST SP 800-30 Rev. 1](#)

Purpose

Databases servers are utilized for storing and collecting both public and private data for usage, analytics, and other needs to strengthen business. That is why you observe and see personalized advertisements and brands specified for each customer is because of this.

With strengths in storing and securing data from servers also comes with weaknesses. Since the database server is public (as stated from the scenario), malicious attackers will hack the

servers (via social engineering, SQL injection, DDOS, or other types of attacks) to access both public and sensitive data for nefarious purposes. If the server was disabled due to this attack, the company's business operations would harm the following equities: reputation, customers, profits, and sustainability to operate in the long-term.

That is why vulnerability assessments are necessary to focus on identifying, assessing, and labeling them based from how severe and risks involved.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
<i>Malicious Hacker</i>	• <i>Reconnaissance</i> • <i>Surveillance</i> • <i>Phishing (i.e. craft a well-written e-mail/message as a phishing attempt to sway an employee)</i>	3	3	9
<i>Employee</i>	• <i>Disrupt critical business operations</i> • <i>Attempts for betrayal</i>	2	3	8
<i>Customer</i>	• <i>Alteration and deletion of critical information</i>	1	3	3
<i>Storage Servers</i>	• <i>Faulty physical components</i> • <i>Inability to upgrade (must upgrade components to keep up with the storage of data)</i>	2	1	3
<i>Software</i>	• <i>Malicious Software installed (Not configured rule from firewall, etc.)</i> • <i>Outdated Software (depreciated, no updates/patches)</i>	3	3	7
<i>Temperature & Humidity Controls</i>	• <i>Ventilated physical server storage (cause damage to hardware components thus reducing performance)</i>	3	2	3

Extreme Environment Events	Disaster-prone location (stormy weather, earthquakes, landslides, etc.)	3	3	7
----------------------------	---	---	---	---

Approach

Risks were measured based from the specific factors as to where I'm from (I'm located at Southeast Asia, so I considered the weather, disasters, etc.). The selection of threat sources and their events, associated with their likelihood of risks and how severe it would affect the company's business operations, were mapped out due to the server's public access by the company.

There may be limitations on how I identify other threat sources but these are the general vulnerabilities that were weighed against towards the adverse impact on daily operational requirements.

Remediation Strategy

The required access controls would be to privatized the database server through authentication, authorization, and auditing to ensure only workers with authorized clearance to access the database server. Administrators must implement the following security controls:

- Adhere to the Principle of least privilege (role-based access control)
- Enable MFA & strong password requirements
- Zero-trust for devices
- Firewall and network configuration for data transits (encrypt data in motion & IP allow-listing to prevent random users from connecting to the database server)

The overall assessment will dramatically improve the overall security results of the system in the long run. By identifying and assessing the threats, it will strategically help the company to focus on fortifying the weaknesses of those sources to allow continuous business operations. Failure to alleviate these sources would negatively affect the company's long-term operations in the long run.